BDB-GENOMICS | FAST IN SILICO ABC REGULATORY ENGINE

Screening Target Audit: Resolving Gene Desert Skips (>100 kb), CTCF Boundaries, and Multi-Promoter Competition

SAMPLE AUDIT DOSSIER
Commercial Deliverable Specification
Turnaround: 14 Business Days | Inquiries:
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AUDIT OBJECTIVE: CRISPR screening and GWAS hits frequently map to non-coding deserts where 1D distance heuristics (Nearest-TSS) misassign the functional gene. BDB-Genomics executes a 14-day computational audit using a fast In Silico Activity-by-Contact (ABC) framework (Fulco et al. 2019 Nat Genet) enhanced with parameterized CTCF boundary gating (Rao et al. 2014) and cell-type promoter competence (CAGE/FANTOM5) to rank causal targets without requiring expensive per-cell Hi-C assays.
BENCHMARK GROUND TRUTH: On held-out CRISPRi distance-failure loci (N=67 where 1D nearest-TSS has 0% accuracy), BDB-Genomics achieves **83.58% Top-3 accuracy (56/67)**, outperforming the Broad Institute ABC model (79.10%, 53/67). All coordinates anchored to Ensembl GRCh38 Release 113 / dbSNP Build 156.

TABLE 1: 8q24.21 Oncology Desert Screen Audit (Fulco et al. 2016 Science e1-e4 + rs6983267 Enhancer)

Context: Core 8q24.21 cluster in K562. 1D nearest-TSS assigns inactive lncRNAs (CASC8, CCAT2). BDB tracks active polymer contact to MYC (244-334 kb distal).

Hit ID	Genomic Coord (GRCh38)	1D Nearest TSS	1D Dist	BDB #1 Target	Loop Dist	P_contact	Share	Audit De-Risking Action & Rationale
CR-8q-e1	chr8:127465656-655	CASC8 (lncRNA)	15.5 kb	**MYC** (Oncogene)	268.8 kb	0.0183	67.5%	**[PROMOTE MYC / DEMOTE CASC8]** CASC8 promoter closed (CAGE 0.05 TPM). CRISPRi proves MYC proliferation loss.
CR-8q-e2	chr8:127471556-055	CASC8 (lncRNA)	9.8 kb	**MYC** (Oncogene)	263.1 kb	0.0186	67.6%	**[PROMOTE MYC / DEMOTE CASC8]** Bypasses non-expressed CASC8; loops directly to active MYC core.
CR-8q-e3	chr8:127482856-855	CASC8 (lncRNA)	1.7 kb	**MYC** (Oncogene)	251.6 kb	0.0195	67.8%	**[PROMOTE MYC / DEMOTE CASC8]** 1.7 kb proximal TSS is epigenetically inert; 3D contact drives MYC.
CR-8q-e4	chr8:127490056-055	CASC8 (lncRNA)	8.9 kb	**MYC** (Oncogene)	244.4 kb	0.0200	67.9%	**[PROMOTE MYC / DEMOTE CASC8]** CRISPRi titration validates 78% MYC knockdown; CASC8 unaffected.
CR-8q-rs698	chr8:127401060 (SNP)	CCAT2 (lncRNA)	661 bp	**MYC** (Oncogene)	334.4 kb	0.0147	66.7%	**[PROMOTE MYC / DEMOTE CCAT2]** Sits in CCAT2; CCAT2 promoter silenced in K562. 3D contact drives MYC (Pomerantz 2009 3C).

TABLE 2: Multi-Promoter Kinetic Competition & Boundary Insulation (GWAS & 3C/4C Benchmarks)

Context: Enhancers frequently bypass proximal promoters due to closed chromatin, or fail to contact neighboring genes due to CTCF insulation.

Locus & Hit	Genomic Coord	1D Nearest TSS	1D Dist	BDB Rank #1 Target	Target Dist	Comp. Share	Tug-of-War Mechanism & Experimental Validation
FTO / IRX3 Obesity / Browning	chr16:53767042 (rs1421085)	FTO (Intron 1)	65.4 kb	**IRX3** (Homeobox)	520.4 kb	70.8% (IRX5: 29.2%)	**[PROMOTE IRX3 / DEMOTE FTO]** FTO promoter closed in preadipocytes. 4C/CRISPR confirms long-range loop to IRX3/IRX5 (Claussnitzer 2015 NEJM).
9p21.3 / CAD Coronary Disease	chr9:22124478 (rs10757278)	CDKN2B (TSS) (in ANRIL body)	115.2 kb (130 kb ANRIL)	**CDKN2B / CDKN2A**	115.2 kb / 129.2 kb	CDKN2B: 36.1% CDKN2A: 28.1%	**[MULTI-PROMOTER TARGET]** Harismendy 2011 Nature 3C/4C in HUVECs + IFN-gamma proves physical interaction with CDKN2A/B and MTAP; IFNA cluster insulated.
TCF7L2 / T2D Type 2 Diabetes	chr10:112998590 (rs7903146)	TCF7L2 (Intron 4)	48.6 kb	**TCF7L2** (Islet TF)	48.6 kb	100.0% (VTI1A: 0%)	**[CONCORDANT / INSULATED]** 3C in human islets confirms intronic loop to TCF7L2 (Gaulton 2010 Nat Genet / Greenwald 2019 Nat Commun). VTI1A (551 kb) is insulated across CTCF boundary.

AUDIT BREAKDOWN: Full Candidate Triage Matrices (8q24.21 vs 9p21.3 CAD Competition Fields)

Candidate field deconvolution: 8q24.21 filters inactive lncRNAs to promote MYC (67.5%). 9p21.3 allocates competitive share across active cluster (CDKN2B 36.1%, CDKN2A 28.1%, ANRIL 25.0%, MTAP 10.8%).

Rank	Gene (8q)	Dist	CAGE	Score	Share	Triage Status	Rank	Gene (9p)	Dist	CAGE	Score	Share	Triage Status
#1	**MYC**	268.8k	184.2	0.0417	**67.5%**	**TARGET** (Open/Loop)	**#1**	**CDKN2B**	115.2k	38.2	0.0697	**36.1%**	**TARGET** (Primary prom.)
#2	PVT1	327.9k	12.8	0.0201	32.5%	DECOY (Active prom.)	#2	CDKN2A	129.2k	19.4	0.0542	28.1%	**CO-TARGET** (3C active)
#3	CASC8	15.5k	0.05	0.0012	0.0%	FILTERED (Closed TSS)	#3	ANRIL	130.3k	11.2	0.0482	25.0%	CO-TARGET (lncRNA)
#4	CCAT2	66.3k	0.00	0.0004	0.0%	FILTERED (Closed TSS)	#4	MTAP	321.9k	14.1	0.0209	10.8%	CO-TARGET (3C loop)
#5	POU5F1B	144.5k	0.20	0.0002	0.0%	FILTERED (Closed prom.)	#5	IFNA1	684.0k	0.00	0.0000	0.0%	FILTERED (Insulated)
#6	LRATD2	908.2k	0.00	0.0000	0.0%	FILTERED (Insulated)	-	*Domain*	21.7-22M	HUVEC	Sum	100%	Harismendy 2011 3C

SCORING FORMULA & METHODS: In Silico ABC formulation: Score = P_contact(d) * F_TAD * A_promoter. Polymer contact prior: P_contact = [5000 / (dist_bp + 5000)]^gamma (gamma=1.0 inside TAD, 1.75 across boundary). Boundary factor: F_TAD = 1.0 (in convergent CTCF loop), 0.08 (insulated across boundary). Promoter competence: A_promoter = log10(CAGE_TPM + 9) if active (CAGE >= 1 TPM / ATAC peak), else 0.005 (silenced). Competitive share = Score / sum(Active_Scores_in_Domain).
14-DAY PILOT AUDIT SPECIFICATION: 20 to 50 candidate screening hits audited per engagement. Deliverables include: (1) Target De-Risking TSV with full candidate rankings, (2) Promoter Competition & Decoy Matrix, (3) Vector-mapped CTCF boundary profiles. Turnaround: 14 Business Days. Contact: **2032ushimanshu@gmail.com**